Apurbo Banik Turjo

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Summary

Highly motivated and results-oriented Software Engineer with **6 months of industry experience**, passionate about **AI** and **software engineering**. Proven ability to design, develop, and deploy robust applications using modern technologies, including **microservices**. Recently completed an undergraduate ML thesis on **dengue breeding site detection**, achieving **83.6% balanced accuracy**, currently under review for publication. Seeking innovative opportunities to leverage expertise in **Java, Spring Boot, React, Docker**, **Nginx, Jenkins**, and **Machine Learning**.

Education

Bangladesh University of Engineering and Technology — Dhaka, Bangladesh
B.Sc. in Computer Science and Technology; ***CGPA: 3.8732/4.00*** — *Feb 2020 – March 2025*
Relevant courseworks: *DSA (I, II), OOP, System Design, Database, OS, ML, Compiler, etc.*
Awards: *Dean's List (2 semesters)*

Shahed Syed Nazrul Islam College — Mymensingh, Bangladesh
Higher Secondary Certificate; ***GPA: 5.00/5.00*** — *2019*

Skills

Programming Languages: C/C++, Java (8, 21), Python, JavaScript, TypeScript, SQL, Assembly, Bash
Frameworks: Spring, Spring Boot, Hibernate, Node, React, Django, Express, Microservices
Tools & Platforms: Git, Github Workflow, Docker, Postman, Figma, n8n, MCP server, Nginx, Jenkins
Databases: PostgreSQL, MySQL, OracleDB
Cloud Computing & Hostings: Azure, AWS, Render
Machine Learning: ML Thesis (under review), Model Compression, Fairness Evaluation, Federated Learning

Experience

Software Engineer — *[Dhaka, Bangladesh]*
Apr, 2025 – Present

- Developed and maintained robust backend services using **Java (8 & 21), Spring, Spring Boot**, and **Hibernate**.
- Implemented a **JWT-based authentication service** with session refresh and public key rotation cycles, ensuring secure and scalable access for other microservices.
- Gained hands-on experience with **React** for building dynamic and responsive user interfaces.
- Successfully **dockerized** a subproject, optimizing build cache usage for efficient containerization.
- Collaborated on various software development projects, contributing to all phases of the SDLC, including experience with **Nginx** for proxying, and **Jenkins** for CI/CD pipelines.

Projects

EduByte [React, Bootstrap, Express, PostgreSQL] **GitHub** | **Video Demonstration**

- EduByte is an e-learning platform offering computer science courses, structural organization of the course contents, interactive AI chat-bot, automated work assessments, and progress tracking.
- Functionalities like notification system, course request, course update, course upload. A discussion forum where students can post, react and comment making collaborative learning more accessible.

Portfolio [React, MaterialUI, Express, Supabase, Figma] | **Visit** | **Github**

- Developed a personal portfolio to showcase projects, photographies and technical skills using React on the frontend and Express on the Backend. Automated the project display process with a responsive design with Material UI.

Football Player Management System [Java, JavaFX, CSS] | **GitHub**

- Player management for clubs, enabling searches, player trades, and reports. It uses multi-threading, Inter-process communication, file-based input/output and operates via JavaFX, CSS based GUI.

FunTimeProject - Weather Web App, Basic Calculator [JS, HTML, API]

- The **Weather-web-app** fetches real-time weather information of the listed capitals and the country flag using raw JS-API. **Visit** | **Github**
- The **BasicCalculator** is built using raw JS, HTML & CSS. **Visit** | **Github**

Research Experience

Model Compression Induced Fairness Evaluation

- Currently researching fairness evaluation in conjunction with model compression and federated learning, exploring critical ethical considerations in AI deployment.

Automatic Dengue Breeding-site Identification using Drone Images [ML, CV, Annotation, Python, Matlab]

- Analyzed dengue breeding sites in dense urban environments using UAV imagery to support proactive public health interventions. Focused on automating potential risky building detection (achieved **83.6% balanced accuracy**) and risk assessment for effective resource allocation.
- Undergraduate thesis project, currently in review for publication.

Hobbies

Reading, Running, Swimming, Fitness, Table Tennis